Chemical Engineering Journal 499 (2024) 155865

Contents lists available at ScienceDirect

Chemical Engineering Journal

journal homepage: www.elsevier.com/locate/cej

Chemical recycling of polyethylene terephthalate using a micro-sized MgO-incorporated SiO_2 catalyst to produce highly pure bis(2-hydroxyethyl) terephthalate in high yield

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ARTICLE INFO

Keywords:
Polyethylene terephthalate
MgO/SiO_2
Glycolysis
Catalysis
Micro-size catalyst

ABSTRACT

Chemical recycling is a feasible method for achieving the closed-loop recycling of polyethylene terephthalate (PET). In this strategy, post-consumer PET is depolymerized to bis(2-hydroxyethyl) terephthalate (BHET) via glycolysis, which is then repolymerized to yield recycled PET (rPET). Herein, we introduce an innovative approach for the chemical recycling of PET that focuses on glycolysis facilitated by a microsized MgO-incorporated SiO_2 catalyst. This choice of catalyst is groundbreaking owing its superior glycolysis-catalysis efficiency and the ease with which it can be separated from the reaction mixture, thereby addressing the common challenges faced by traditional catalysts during PET depolymerization. By employing a heterogeneous catalyst with significantly larger particles than a conventional catalyst, we not only aimed to simplify the catalyst-removal process, thereby improving BHET purity, but also observed enhanced efficiency during the catalytic reaction that led to a superior yield of the desired product. The catalyst was synthesized using a wet impregnation method and its efficiency was evaluated at various reaction times and dosages, which revealed that the microsized MgO/SiO_2 catalyst significantly outperforms traditional catalysts, to deliver a 95.1% yield of BHET with negligible amounts of metal detected in the final product. This study not only demonstrated the potential of microsized catalysts during PET glycolysis, but also contributes to the advancement of sustainable PET-recycling practices. We anticipate that the microsized MgO/SiO_2 catalyst is a promising alternative for the industrial upcycling of PET waste.

1. Introduction

Plastics is a collective term for synthetic or semi-synthetic materials intentionally made by humans that use polymers as the main material. Since they were first synthesized in 1869, plastics have become industrially important owing to their high production levels, low unit costs, high durability, and chemical resistance. A total of 390.7 million tons of plastic were produced globally in 2021, with 352.3 million tons (i.e., 90.17 %) being fossil-based [1]. Polyethylene terephthalate (PET), one of the most widely used polyester plastics, is a versatile thermoplastic polymer used to manufacture bottles, textiles, packaging, and film materials owing to its excellent chemical, thermal, and mechanical properties and cost-effectiveness [2,3]. The annual production of PET exceeded 30 million tons in 2017 [4] and global demand for PET packaging alone is expected to reach 27.1 million tons in 2025 [5]. Plastics pose a serious environmental threat because their consumption appears to be increasing exponentially and most plastics decompose naturally with difficulty [6]. Landfilled PET waste photodegrades into smaller fragments over time, leading to microplastics that accumulate in the marine environment and are eventually ingested and accumulated by living organisms, including humans [7]. Approximately 46 % of plastic waste is incinerated or dumped into the environment [8–11]; consequently, developing efficient methods for increasing their recycling rate is important.

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https://doi.org/10.1016/j.cej.2024.155865
Received 11 June 2024; Received in revised form 11 September 2024; Accepted 15 September 2024
Available online 16 September 2024

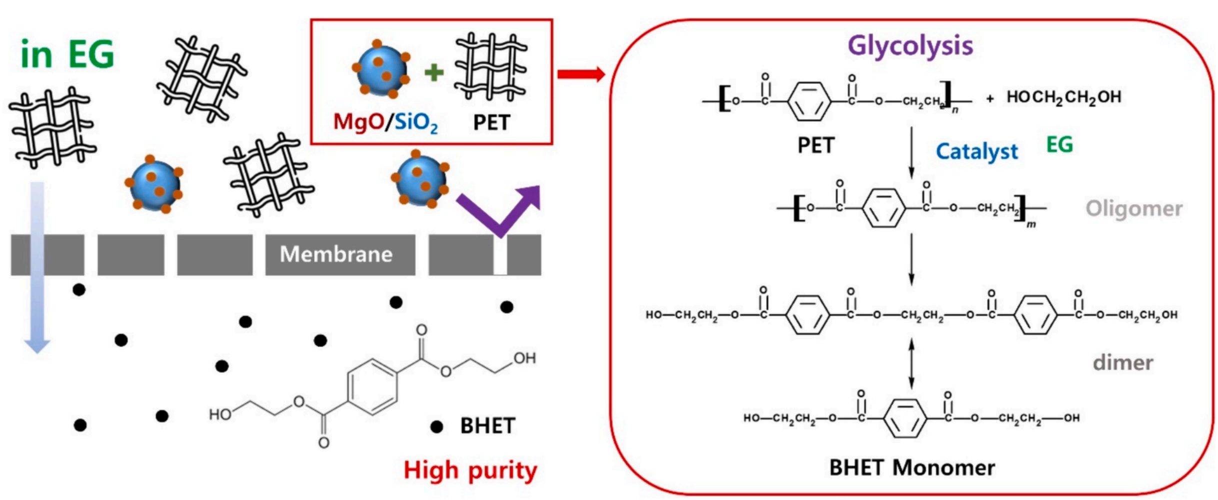
To date, considerable effort has been devoted to investigating recycling processes, with chemical recycling considered to be a promising and sustainable solution among various reported strategies [12]. Chemical recycling involves depolymerization to yield monomers or oligomers that can be reused as high-purity starting materials. Chemical recycling contributes to the development of a circular economy by valorizing polymers into high-value monomers via chemical depolymerization. PET-waste chemical recycling processes include methanolysis, hydrolysis, glycolysis, ammonolysis, and hydrogenation [13], which all involve solvolytic chain cleavage. Among these, glycolysis an accepted commercial method owing to its mild reaction conditions, low solvent volatility, and high monomer quality. Glycolysis involves transesterification of the ester groups in PET with ethylene glycol (EG) to form bis(2-hydroxyethyl) terephthalate (BHET), which can be isolated by crystallization and used as a raw material for the preparation of repolymerized PET (rPET)[14]. However, glycolysis requires a catalyst because it is an extremely slow reaction [15]; consequently various PET-glycolysis catalysts have been developed to enhance reaction efficiency, including metal oxides [16,17], ionic liquids [18], urea [19] and enzymes [20]. However, the relatively high temperatures associated with glycolysis also lead to catalyst decomposition, which affects the quality of the rPET following repolymerization. Indeed, high levels of catalyst residues (which are monomer impurities) present in the BHET crystals following crystallization must be removed prior to polymerization in order to produce rPET of similar quality to virgin PET from the depolymerized monomer [21]. Effort has recently been directed toward addressing the abovementioned issue, with metal-free glycolysis offering an attractive approach because it does not produce metal residues. Zhang et al. developed a phosphazene-based catalyst that delivered 100 % PET conversion with a BHET yield of around 92.7 % [22]. Wang et al. showed that urea can be utilized as a glycolysis catalyst to deliver BHET in 78 % yield under mild temperate conditions [19]. Yue et al. investigated ionic liquids for glycolysis and reported a 71.2 % yield of BHET using [Bmim]OH [23]. However, these metal-free catalytic systems are relatively inefficient and poorly thermostable compared to metal-based catalysts, and ionic liquids are expensive. Magnetic separation offers another strategy for removing residual catalysts. Cano et al. described the use of magnetically recoverable Fe_3O_4@SiO_2@(methylimidazolium)[$FeCl_4$] nanoparticles as a PET-glycolysis catalyst [24]. These researchers incorporated ionic liquids onto magnetic particles that are then used as a magnetically recoverable nano-catalyst that delivered a yield of almost 100 % and was able to be reused for up to 15 times. However, prolonged decomposition (24 h) precluded the use of the ionic-liquid-SiO_2@Fe_3O_4 catalyst. Guo et al. prepared nanosized Mg–Al double oxides sintered onto Fe_3O_4 magnetic microparticles, which exhibited excellent magnetic properties [25]. Magnetic particles are easily retrieved by magnetic decantation and used repetitively. A monomer yield or more than 80 mol% was achieved within 90 min at 240°C using the Mg-Al-Fe_3O_4 catalyst; however, such a catalyst does not separate efficiently when used in large quantities and requires relatively high temperatures, which are drawbacks. Therefore, the development of efficient, stable, and easy-to-separate catalysts for PET glycolysis is necessary.

In this study, we prepared a silica-particle-based micro-sized heterogeneous metal catalyst that is easily separated from the reaction medium. Most literature reports use nanocatalysts to increase catalytic efficiency; however, they can be adsorbed by BHET crystals during crystallization. The main objective of the current study was to improve the conversion efficiency during PET glycolysis using a micro-sized heterogeneous metal catalyst and increase monomer purity by facilitating catalyst removal (Scheme 1). Silica-based catalysts have attracted much attention owing to their unique properties, including their chemical and thermal stabilities, non-toxicities, economic viabilities, and simple preparation methods [26]. Therefore, silica-based catalysts are expected to prevent the formation of residues during high-temperature decomposition reactions. Micro-sized catalysts are particularly important because they are recyclable and easy to separate. We introduced MgO/SiO_2 as a micro-sized metal/silica catalyst for the depolymerization of PET and optimized the reaction conditions for PET glycolysis. Because the catalyst contains large particles, it is easily separated from the reaction medium, thereby enabling very-high-purity BHET and high conversion rates that are comparable to those reported for nano-sized catalysts. In this study, we investigated the catalytic efficiencies of metal oxides supported by micro-sized silica particles for PET degradation and the purity of the produced BHET. This catalyst delivered a 95.1 % yield of negligibly metal-contaminated BHET.

2. Experimental section

2.1. Materials

BHET (≥94.5 % purity (gas chromatography area%), for reference) and magnesium nitrate hexahydrate ($Mg(NO_3)_2{\cdot}6H_2O$) was purchased from Sigma–Aldrich. SiO_2 (Q-15, mean pore diameter: 15 nm; surface area: 200 m^2/g) was supplied by Fuji Silysia, Ltd. (Japan). PET fiber flakes were supplied by Hankuk Ace Co., Ltd. (Republic of Korea). Ethylene glycol (Reagent Plus > 99.9 %) was purchased from Sigma–Aldrich.



Scheme 1. Schematic depicting the depolymerization of PET using the micro-sized heterogeneous MgO/SiO_2 catalyst.

2.2. Characterization

Scanning electron microscopy (SEM, SU8010, Hitachi, Japan) augmented with energy dispersive spectroscopy (EDS) was used to analyze the synthesized catalysts, characterize their morphologies and particle sizes, and confirm their compositions, with five different samples subjected to EDS. X-ray diffractometry (XRD, X'Pert PRO, PANalytical, Netherlands) was used to determine catalyst crystallinity using Cu Kα radiation and a Pixcel 1D detector. Diffraction patterns were collected in the 5–90 2θ range at 3°/min. Brunauer–Emmett–Teller (BET) analysis was used to determine the relationship between the amount of MgO and the specific surface area of the catalyst. Inductively coupled plasma–optical emission spectroscopy (ICP-OES; Ultima Expert, Horiba Scientific, Japan) was used to determine the general amount of Mg in the catalyst. The synthesized BHET was analyzed by Fourier-transmission infrared (FT-IR) spectroscopy (Nicolet iS 50, Thermo Scientific, USA) in the 4000–400 cm^{-1} wavenumber range to confirm that BHET had been formed. ^{1}H NMR spectroscopy (Ascend^{TM} 400, Bruker, USA) was performed at 400 Hz to ensure that the glycolysis product was pure BHET monomer. High performance liquid chromatography (HPLC, Waters, Alliance 2695 / 2996 DAD, USA) with a C18 column (250 mm × 4.6 mm, 5 μm, Waters, USA) was used to determine the purity of the synthesized BHET, with 70:30 (v/v) methanol/water used as the mobile phase at a flow rate of 1 mL/min and 40°C, with an injection volume of 1 μL and a running time of 30 min. A 254-nm wavelength UVD detector was used for analysis. Thermogravimetric analysis (TGA, Auto TGA Q500 thermal analysis system, TA Instrument Inc., USA) was conducted in the room-temperature to 800 °C range to determine the thermal stability of BHET. BHET purity was determined using inductively coupled–plasma mass spectrometry (ICP-MS, Elan 6100, Perkin Elmer, USA), which involved dissolving the catalyst in high-temperature argon plasma at 6000 K with the radio frequency (RF) generator set to 40 MHz, and power levels of 0.5–1.6 kW used. Temperature-programmed desorption (TPD) was performed using an Autochem II 2920 instrument (Micromeritics, USA).

2.3. Synthesis of the MgO/SiO_2 catalyst

MgO/SiO_2 was synthesized using a wet impregnation method. Magnesium precursor solutions with magnesium contents of 2.5, 5, 10, and 30 wt% were prepared by dissolving 2.64, 5.27, 10.55, and 31.65 g magnesium nitrate hexahydrate, respectively, in 100 mL of distilled water. Each solution was stirred for 5 min before adding 10 g of SiO_2. The resulting mixtures were then stirred continuously and heated in an oil bath at 120°C for 12 h to evaporate the solvent. The samples were subsequently calcined at 600 °C for 6 h.

2.4. PET depolymerization via glycolysis

PET fiber (5 g) and EG (50 g) were combined and transferred to a 250-mL three-necked round-bottom flask. The desired amount of catalyst (0.01, 0.02, 0.03, 0.04, or 0.05 g) was then added to the mixture, and the temperature was increased to 196 °C with a heating mantle, at which point glycolysis was allowed to proceed for 1–3 h, with yield measured in 30-min intervals. Heating was discontinued at the end of the reaction, and the system was allowed to cool to 120 °C. Simultaneously, 200 mL of distilled water was heated to 60 °C. The cooled reaction mixture was vacuum filtered to separate the solid catalyst and unreacted PET from the liquid phase containing the glycolysis products, and the filtrate was then mixed with the preheated distilled water and stirred continuously for 30 min. This step aimed to separate BHET monomer from dimers and oligomers based on their solubility differences. The dimers and oligomers crystallized owing to the relatively low temperature, which enabled them to be separated from the monomer. The crystallized dimer and oligomers were collected by filtration, leaving only the BHET monomer in the filtrate. The BHET monomer solution was then allowed to crystallize in a refrigerator at 4 °C for 12 h. The crystallized BHET was collected by filtration and dried in an oven at 90 °C for 4 h. The weight of the BHET crystals was measured to preliminarily determine the yield of the depolymerization process. The calculation formula for the yield of product BHET is as follows:

$$\text{Yield of BHET in residue} = \frac{W_{BHET}/M_{BHET}}{W_{PET}/M_{PET}} \times 100\%,$$

where W_{PET} is the initial weight of the PET, W_{BHET} is the weight of the BHET, M_{PET} is the molecular mass of the PET repeat unit (192 g/mol), and M_{BHET} is the molecular mass of the BHET (254 g/mol).

Furthermore, the filtrate solution was analyzed by HPLC to quantify dissolved BHET. A calibration curve was constructed using the areas of the HPLC peaks of commercial BHET solutions that correspond to yields of 0 %, 20 %, 40 %, and 60 %. (**Fig. S1**) The concentration of BHET in the filtrate solution was calculated by interpolating the corresponding area onto the calibration curve; this concentration was converted into the yield of BHET in the filtrate. The total yield of the BHET product was calculated using the following formula:

$$\text{Total yield of BHET} = \text{Yield of BHET}_{in\ residue} + \text{Yield of BHET}_{in\ filtrate}$$

3. Results and discussion

3.1. Synthesis and characterization of the MgO/SiO_2 catalyst

The primary goal of this study was to obtain highly pure BHET monomer through PET glycolysis while simultaneously enhancing the efficiency of the reaction. To achieve this objective, we synthesized a heterogeneous catalyst with particles greater than 100 μm in size to facilitate easy separation via filtration. To that end, we synthesized MgO/SiO_2 catalysts using a simple wet impregnation method. These micro-sized-silica-supported catalysts were impregnated with varying amounts of Mg in an aqueous medium, after which they were analyzed using several methods to examine their efficacies.

The MgO/SiO_2 catalysts were subjected to SEM, the results of which are shown Fig. 1, which reveals spherical particles larger than 100 μm in size, consistent with our requirements. Despite the MgO content varying between 2.5 and 30 wt%, the various catalysts exhibited consistent particle sizes of ∼ 220 μm, as shown in **Fig. S2**, which indicates that the MgO content does not significantly influence catalyst-particle size, most likely because supporting SiO_2 particles of uniform size were used. Noteworthily, the 30 wt% MgO/SiO_2 catalyst exhibited a distinctly different morphology and structure. SEM images of the catalysts at different magnifications presented in Fig. 1 and S3 clearly reveal that the 30 wt% MgO/SiO_2 particles are cracked and broken, with morphology also observed to depend on the MgO content. The presence of MgO is clearly observable in the image of the 10 wt% MgO/SiO_2 sample, but is more pronounced in 30 wt% MgO/SiO_2.

The Mg content of the MgO/SiO_2 catalysts was further investigated by EDS (**Fig. S4**), which revealed Mg contents of 2.8 ± 1.0, 4.2 ± 0.29, 9.6 ± 3.1, and 21.3 ± 1.6 wt% for the MgO/SiO_2 catalysts nominal 2.5, 5, 10, and 30 wt% loadings, respectively, which reveals a consistent relationship between the amount of metal precursor used and the observed metal content.

The internal structures of the catalysts were explored in more detail using cross-sectional SEM and EDS (**Fig. S5**). For this purpose, the 10 wt % MgO/SiO_2 catalyst sample was crushed to expose its cross-section, thereby facilitating detailed observation. The cross-sectional image clearly shows a distinct MgO layer on each SiO_2 particle. Additionally, the regions corresponding to broken SiO_2 particles and boundary areas with MgO were subjected to EDS mapping, which confirmed the presence of magnesium and oxygen in the expected ratios, consistent with layers composed of MgO. These findings confirm that MgO is present on the SiO_2 surface. The detailed SEM images and EDS maps shown in Fig. S4 provide visual evidence of the structures and compositions of the

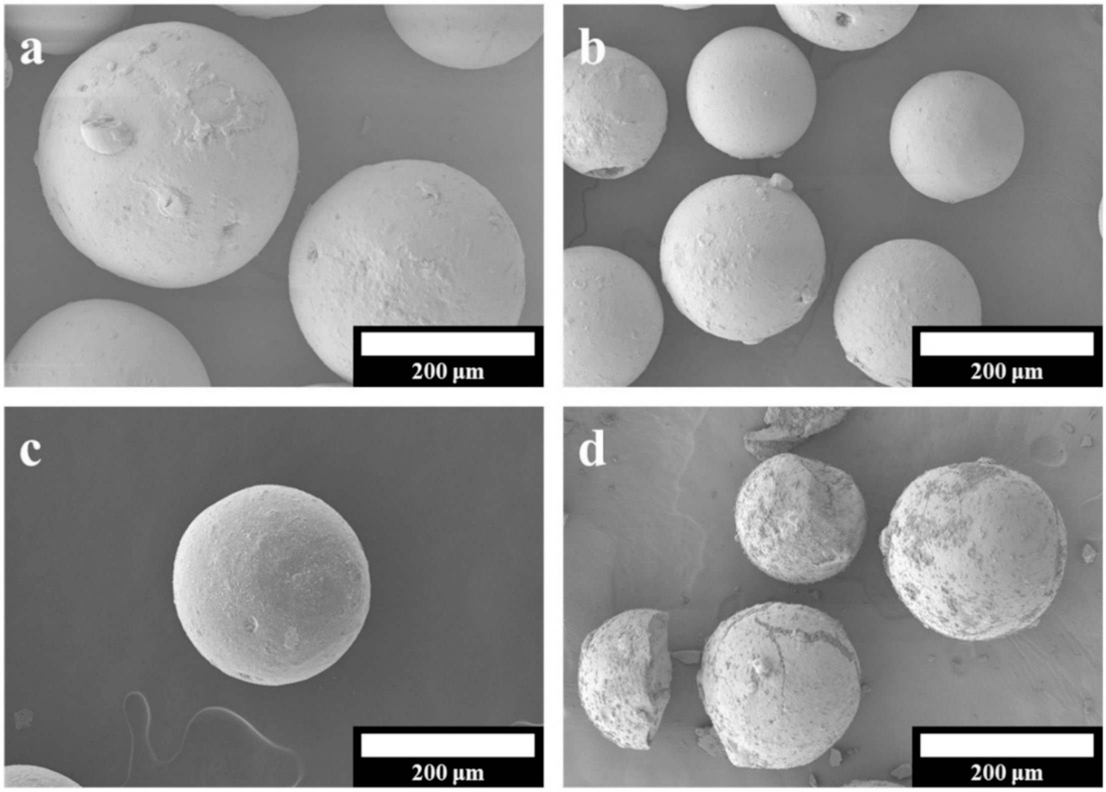


Fig. 1. SEM images of the: a) 2.5, b) 5, c) 10, and d) 30 wt% MgO/SiO_2 catalysts.

catalysts.

Furthermore, catalyst surfaces were analyzed using the Brunauer–Emmett–Teller (BET) method, which revealed that the surface area decreases with increasing metal content, from 236.89 m^2/g at 2.5 wt% to 128.61 m^2/g at 30 wt%. We conclude that the incorporation of MgO into the SiO_2 matrix has a pronounced impact on the pore structure of the catalyst, as evidenced by the observed decrease in surface area with increasing MgO content. This observation is attributable to the MgO particles clogging the silica pores. The results presented above are summarized in Table 1 to provide a clearer overview of the relationship between MgO content and the physical properties of the catalyst.

We used X-ray diffractometry (XRD) to further investigate the structural characteristics of the MgO/SiO_2 catalysts and provide deeper insight into the crystalline nature of MgO/SiO_2 (**Fig. S6**). Each catalyst displays a prominent peak at 21.8° characteristic of amorphous SiO_2 and indicative of the silica support structure [27]. Notably, only the 30 wt% MgO/SiO_2 sample exhibited clear crystalline MgO peaks, whereas the other samples (2.5, 5, and 10 wt% MgO/SiO_2) did not exhibit distinct peaks. The XRD pattern of the 30 wt% MgO/SiO_2 sample shows peaks at 32.4°, 36.6°, 42.9°, and 62.4° that correspond to MgO, and at 35.6°, 39.9°, and 52.4° that correspond to Mg_2SiO_4, indicative of a significant MgO crystalline phase that interacts with the silica support [28,29]. With the exception of the peak corresponding to silica particles, the other samples did not show any XRD peaks. The absence of these MgO-related peaks in the samples with lower MgO contents implies the presence of less-pronounced crystalline structures, which is likely due to their lower Mg contents and their distribution within the silica matrices. The clear peaks observed for the 30 wt% MgO/SiO_2 sample indicate the presence of MgO, confirming its successful incorporation onto the silica support. However, while the 30 wt% MgO/SiO_2 sample demonstrated significant crystalline MgO incorporation, it also exhibited a disrupted morphology and a smaller surface area, which can potentially affect the purity of the monomer. Conversely, catalysts with lower MgO contents maintained structural integrity but lacked the distinct crystalline MgO phases essential for catalytic activity. We conclude that the 10 wt% MgO/SiO_2 catalyst is the optimal candidate by considering the balance between structural integrity, surface characteristics, and the presence of crystalline MgO.

Table 1
Structural Properties of MgO/SiO_2 Catalysts with Different Mg Contents.

	BET (m^2/g)[a]	Mg (wt%)[b]	Si (wt%)[b]
2.5 wt%	236.89	2.8 ± 1.0	38.6 ± 2.0
5 wt%	200.8	4.2 ± 0.29	36.5 ± 1.3
10 wt%	185.2	9.6 ± 3.1	32.7 ± 3.5
30 wt%	128.61	21.3 ± 1.6	24.0 ± 1.7

[a] BET surface area determined from the N_2-isotherm. [b] Metal content determined by multipoint EDS.

3.2. *Catalytic PET-glycolysis performance of Mg/SiO_2*

Catalytic efficiency during the depolymerization of PET is influenced by a range of factors, including the intrinsic properties of the catalyst, its interactions with the reactant, and operational parameters, such as reaction duration, temperature, and catalyst dosage. This study assessed the performance of various MgO/SiO_2 catalysts by examining BHET yields across various reaction times and catalyst dosages to determine the optimal conditions. Our approach established the ideal Mg content for the MgO/SiO_2 catalyst, along with the optimal reaction time and catalyst dosage. Given that magnesium-based catalysts are widely recognized to be effective for PET glycolysis [30], we specifically impregnated silica particles that are approximately 200 μm in size, with magnesium. Silica-particle size was determined by their ability to enhance the purity of BHET by facilitating catalyst removal. The optimal Mg content was investigated to enhance PET-glycolysis efficiency by adjusting the reaction time and catalyst dosage. The effect of Mg content on catalytic activity was assessed by comparing performance at different Mg contents using catalyst dosages of 0.02 and 0.05 g (**Fig. S7**). The both catalyst dosages showed clear trends in which higher BHET yields were obtained at higher Mg contents, which indicates that a higher Mg content expedites the reaction, leading to earlier yield saturation. To confirm the reason for this activity change, TPD analyses were conducted (**Fig. S8** and **Table S1**). The CO_2-TPD results reveal that as the amount of the basic catalyst MgO increased, the adsorption amount of CO_2 also increased. This result is consistent with the observed activity change, as the increased number of MgO active sites enhances CO_2 adsorption, leading to the activity change. The MgO/SiO_2 sites are basic, which suggests that glycolysis follows a conventional reaction mechanism, as reported in various studies [31–34]. The reaction mechanism involves electrophilic attack by the active MgO sites to form bonds with the ester carbonyl oxygens in PET via Lewis acidic sites. Another bond is subsequently formed through the nucleophilic attack of a hydroxyl oxygen of EG at the carbonyl carbon on the opposite side of the ester bond, which breaks the PET hydrocarbon chain. PET becomes disconnected through catalyst desorption, and EG is substituted at both ends of the terephthalate, which sequentially decomposes PET into an oligomer, dimer, and BHET monomer. The corresponding reaction mechanism is shown in **Fig. S9**. Therefore, CO_2-TPD revealed that the basic MgO/SiO_2 site of the catalyst acts as the active site in the PET-glycolysis reaction, consistent with the results of previous studies [35].

Furthermore, an increase in the quantity of the catalyst directly improved the efficiency of the reaction, as expected. A catalyst dosage of 0.02 g resulted in diminished glycolytic activity (~20 %) at the lowest Mg content of 2.5 wt%, whereas a dosage of 0.05 g showed improved PET decomposition (~60 %). However, despite this improvement, a 0.05 g dosage of a catalyst with 2.5 wt% Mg exhibited inferior performance compared to that obtained at higher Mg contents. A dosage of 0.05 g consistently led to substantially high BHET yields (>80 %) for all examined magnesium contents, with the exception of the lowest Mg content of 2.5 wt% (~60 %), which, although improved, is still less efficient that the catalysts with higher Mg contents. We conclude that 10 wt% Mg is sufficient for glycolysis given that both materials are highly catalytically efficient, and that the catalyst is structurally stable. Notably, the catalyst with 10 wt% Mg exhibited superior activity, with peak yield achieved in just one hour. This optimal Mg content was further corroborated by examining the reaction time and loading of the 10 wt% MgO/SiO_2 catalyst.

Table 2 lists BHET yields obtained using the 10 wt% MgO/SiO_2 catalyst at various reaction times and dosages, which are depicted graphically in Fig. 2. The results demonstrate a clear relationship between catalyst dosage and the time required to achieve a BHET yield above 90 %. A higher catalyst loading reduces the time required to obtain yields above 90 %. Specifically, a catalyst dosage of 0.05 g delivered a yield greater than 90 % within 1 h, while 0.04 g required 1.5 h, 0.03 g required 2 h, and 0.02 g required 2.5 h. A lower dosage (0.01 g) did not deliver this level of yield within the 3-h experimental period. These results suggest that higher catalyst dosages accelerate the reaction rate, thereby more-rapidly delivering yields above 90 %. Additionally,

Table 2
Total yields of BHET obtained using 10 wt% MgO/SiO_2.

Time (h)	Total BHET yield (%) 0.01 g[a]	0.02 g[a]	0.03 g[a]	0.04 g[a]	0.05 g[a]
1	20.6 ± 5.52	33.8 ± 5.41	80.2 ± 5.35	79.9 ± 5.27	90.4 ± 5.19
1.5	39.0 ± 3.47	77.4 ± 3.56	82.2 ± 3.26	91.8 ± 3.37	91.0 ± 3.46
2	60.2 ± 1.24	81.7 ± 1.15	95.1 ± 0.97	92.5 ± 0.95	94.6 ± 1.05
2.5	72.2 ± 0.86	92.1 ± 0.74	93.2 ± 0.65	92.0 ± 0.76	93.1 ± 0.88
3	79.8 ± 0.38	93.0 ± 0.43	92.4 ± 0.32	93.1 ± 0.28	93.4 ± 0.33

[a] Catalyst dosage.

we showed that the standard deviation of the yield decreases with increasing time, which indicates that an extended reaction time leads to improved saturation. These findings highlight the importance of catalyst dosage and reaction time in enhancing the efficiency and cost-effectiveness of PET glycolysis. A particularly important aspect of our study is the observation that a high yield can be achieved using a minimal amount of catalyst within a short reaction time, which significantly surpasses the results of other catalysts reported in the literature.

We also investigated the effect of reaction temperature on the glycolysis reaction using 0.05 g of the 10 wt% MgO/SiO_2 catalyst, the results of which are shown in **Fig. S10**. As anticipated, higher temperatures and longer reaction times led to higher reaction yields, which is consistent with the trends shown in Fig. 2. Notably, yields of approximately 95 %, 90 %, 85 %, and 80 % were obtained at 190, 185, 180, and 175 °C, respectively, although progressively longer reaction times were required. These results highlight the efficacy of the catalytic system developed in this study, and demonstrate that high efficiency can be achieved under relatively mild conditions, thereby outperforming other glycolysis systems reported in the literature [36–38]. Furthermore, reactions at lower temperatures offer significant benefits in terms of energy consumption, which is a critical consideration for industrial applications and the broader context of glycolysis research. The catalyst developed in this study not only facilitates rapid depolymerization at elevated temperatures but is also more than 90 % efficient at around 180 °C, which underscores the versatility and practical importance of our catalyst, and provides a valuable solution that balances energy efficiency with high reaction yields.

Table 3 provides a comparative summary of BHET yields based on the catalyst, reaction time, catalyst-to-PET ratio, temperature, and particle size, drawing on recent publications in this field. The use of 10 wt% MgO/SiO_2 catalyst resulted in a yield of 95.1 % within 2 h, despite the relatively low catalyst ratio of 0.6 wt%. Notably, this yield surpasses most of those previously reported. Moreover, high yields were obtained using lower quantities of catalyst. Additionally, this study demonstrated remarkable catalytic efficiency, even when particles larger than conventional were used. The enhanced performance of the micron-sized catalyst is attributable to its unique heterogeneous structure, which engenders it with a high specific surface area and reaction contact area. The incorporation of MgO on the silica surface facilitates glycolysis and ensures consistent catalytic activity despite the use of large particles. Consequently, this study successfully delivered substantially higher yields in shorter duration times using lower catalyst quantities.

In addition, we investigated the recyclability of the catalyst by evaluating its structural stability and catalytic performance multiple reaction cycles. First, a comparison of the catalyst before and after the reaction using SEM imaging (**Fig. S11**) confirmed that the catalyst retained its original morphology, indicating that its structural integrity remained intact throughout the process. This stability suggests that the catalyst does not undergo significant morphological changes, which could otherwise affect its performance in subsequent cycles. To further assess the catalyst's recyclability, we conducted a series of reactions using the recycled catalyst under identical reaction conditions: 0.03 g of catalyst, 196 °C, 2 h of reaction time, with 5 g of PET and 50 g of EG. After recycling, the BHET yield remained consistent at 92.28 ± 1.19 %, which is slightly lower than the initial yield of 95.14 ± 0.97 %, but still demonstrates the catalyst's strong performance even after the first use. The slight reduction in yield is likely attributed to inevitable losses or damage to the catalyst during the recovery process, such as catalyst being trapped in the membrane or damage occurring while scraping the membrane. Despite this, the results emphasize the strong potential for catalyst recovery and reuse, illustrating the practical benefits of the micro-sized catalyst. These findings highlight the significance of the catalyst's recyclability, presenting it as a promising solution for enhancing the sustainability of PET recycling processes.

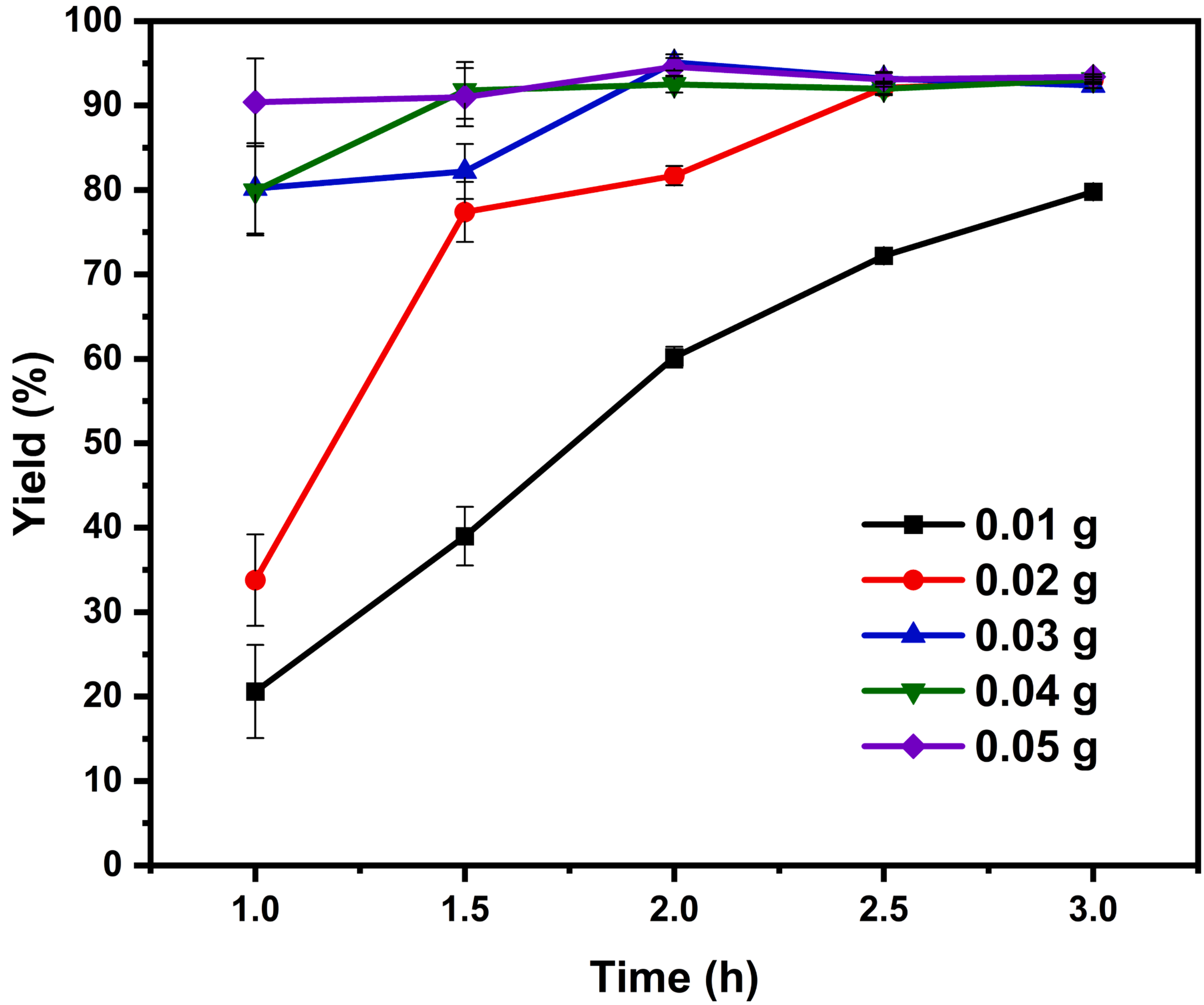


Fig. 2. PET-glycolysis performance (BHT yields) of the 10 wt% MgO/SiO_2 catalyst at different loadings as functions of reaction times.

Table 3
Summarizing PET-glycolysis results reported in the literature.

Catalyst	Time(h)	Catalyst-to-PET Ratio	Particle Size	Temp(°C)	$Yield_{BHET}$ (%)	Ref.
10 wt% MgO/SiO_2	2	0.6 (wt%)	~200 μm	196	95.1	This work
Si-TEA	1.7	15.5 (mol%)	40–63 μm	190	88.5	[36]
CeO_2	0.25	1 (wt%)	2.7 nm	196	90.3	[39]
$CoFe_2O_4$	6	1.33 (wt%)	~20 μm	190	83.0	[40]
Zinc titanate nanotubes	3	0.33 (wt%)	–	196	87	[41]
$WZn_3(H_2O)_2(ZnW_9O_{34})_2$	40 (min)	0.5 (wt%)	–	190	84.5	[37]
ZIF-8	1.5	1 (wt%)	–	197	76.75	[42]
Fe_3O_4-boosted MWCNT	2	5 (wt%)	2.97 nm	190	100	[17]
γ-Fe_2O_3	1	5 (wt%)	10.5 nm	300	90	[43]
SO_4^{2-}/Co-Zn-O	3	0.3 (wt%)	10 μm	180	75	[38]
GO-Mn_3O_4	80 (min)	1 (wt%)	–	300	96.4	[44]
Mg-Al-O@Fe_3O_4	1.5	0.5 (wt%)	10–50 μm	240	80	[25]

3.3. Purity of the BHET monomer derived from glycolysis using the MgO/SiO_2 catalyst

The purity of the BHET derived from the glycolysis of polyethylene terephthalate (PET) is crucial for the recycling process as it impacts the quality of the resulting polymeric material [45]. High-purity BHET is essential for preserving the integrity of the polymerization process, which directly affects the mechanical and physical properties of the recycled product. Impurities in the BHET can compromise these critical attributes, resulting in rPET products of inferior quality. Pawlak et al.

established standards for the physical and chemical properties of PET, which state that the water content must not exceed 0.5 wt% and metal impurities cannot exceed 3 ppm. Exceeding these thresholds can seriously affect the properties of the rPET; for example, an impurity level of 4.3 wt% resulted in 44 % lower PET viscosity [46]. In another study, Torres et al. investigated the properties of rPETs from scrap contaminated with substances such as PVC and adhesives and compared them to those of virgin PET [47], which revealed a dramatic lower elongation at break, with virgin PET exhibiting 270 % elongation, while rPETs containing 20–6000 ppm PVC exhibiting elongations as low as just 5 %, highlighting the importance of highly pure BHET for sustainable PET recycling. The following analysis focuses on assessing the purity of the BHET monomer produced through MgO/SiO_2-catalyzed glycolysis.

We used TGA to investigate the thermal behavior of both the monomer and polymer to assess the purity of the BHET derived from the glycolysis reaction. TGA provides insight by comparing the thermal decomposition profiles of the reference BHET, produced BHET, and raw PET polymer (Fig. 3**a**). The raw PET exhibited distinct thermal decomposition at 410 °C, which is characteristic of PET degradation. In contrast, the TGA curves for both the reference and produced BHET reveal two major weight-loss events: an initial decomposition peak at approximately 210 °C that is attributable to the breakdown of BHET, and a subsequent peak at 410 °C that is likely to correspond to the decomposition of residual PET, which suggests that repolymerization may occur at elevated temperatures during TGA [48]. Therefore, the overlapping thermal-decomposition peaks associated with BHET and any residual PET complicates determining purity. Nevertheless, comparative analyses of the thermal profiles of commercially available BHET and the BHET produced in our study provide crucial information. Both sets of TGA curves exhibit similar thermal behavior, particularly in relation to the characteristic decomposition peaks at 240 and 410 °C; the produced BHET exhibits very similar thermal behavior to that of commercial BHET, with no significant variations in thermal characteristics observed. Although not a definitive measure of purity, this comparison supports the conclusion that the synthesized BHET possesses thermal characteristics comparable to those of commercial BHET.

We used FT-IR spectroscopy to complement the TGA results and further verify the chemical structure of the BHET produced by PET glycolysis (Fig. 3**b**) The FT-IR spectrum of the produced BHET closely resembles that of commercial BHET, indicative of successful depolymerization. Significant peaks that correspond to the characteristic functional groups of BHET are observed, including a broad absorption band around 3271 cm^{-1} and a sharp peak at 1712 cm^{-1} [49]. The presence of the benzene ring, a fundamental component of BHET, is confirmed through peaks at 724 cm^{-1} and in the 1456–1505 cm^{-1} range, with ether linkages evidenced by peaks at 1267 and 1245 cm^{-1} [50].

The findings presented above suggest that the chemical structure of the produced BHET closely mirrors that of commercially available BHET, which is further supported by their similar FT-IR spectra, particularly in the characteristic regions. The main difference between the spectra of the commercial and produced BHET is observed at 1690 cm^{-1}, where two distinct characteristic carbonyl peaks are observed at 1690 and 1712 cm^{-1} for the former. The peak at 1690 cm^{-1} in the spectrum of commercial BHET is suggestive of the presence of an additional material that we assume is the terephthalic acid (TPA) monomer, as the literature reports that TPA absorbs at 1690 cm^{-1} [51]. This suggests that the BHET produced using our glycolysis method is of higher purity than the commercial material, as corroborated by other experiments. Consequently, this analysis provides additional evidence that the MgO/SiO_2-catalyzed glycolysis process effectively yields a BHET monomer with a quality comparable to that of an established commercial material.

While FT-IR spectroscopy and TGA provided valuable information regarding the formation of BHET, they offer limited insight into its purity; hence, we subjected the BHET produced through glycolysis and commercial BHET to HPLC and compared their chromatograms, the results of which are shown in Fig. 4. This method more precisely quantifies the purity of the BHET dissolved in ethylene glycol. The HPLC profile of commercial BHET shows two prominent peaks at retention times of 3.00 and 4.64 min that corresponding to the BHET monomer and its dimer, respectively, with an additional smaller peak observed at 9.6 min that is likely to correspond to an oligomer. The absence of additional peaks in the profile of the produced BHET suggests that oligomers and other impurities are absent and demonstrates that the produced BHET contains lower levels of oligomers and impurities compared to the commercial standard, highlighting its superior purity. Consequently, the HPLC results confirm the successful production of BHET and highlight its superior purity, as evidenced by the absence of dimers and oligomers, which sets the stage for advanced PET recycling.

^{1}H NMR spectroscopy provided detailed regarding the chemical structure of BHET (Fig. 5). Accordingly, we compared the spectra of commercial BHET and a known BHET dimer. Peak 4 (at 8.12 ppm in Fig. 5) corresponds to the aromatic protons of BHET, whereas peak 1 at 4.94 ppm corresponds to a hydroxyl proton, and peaks 2 and 3, at 3.73 and 4.33 ppm respectively, are attributable to the methylene protons

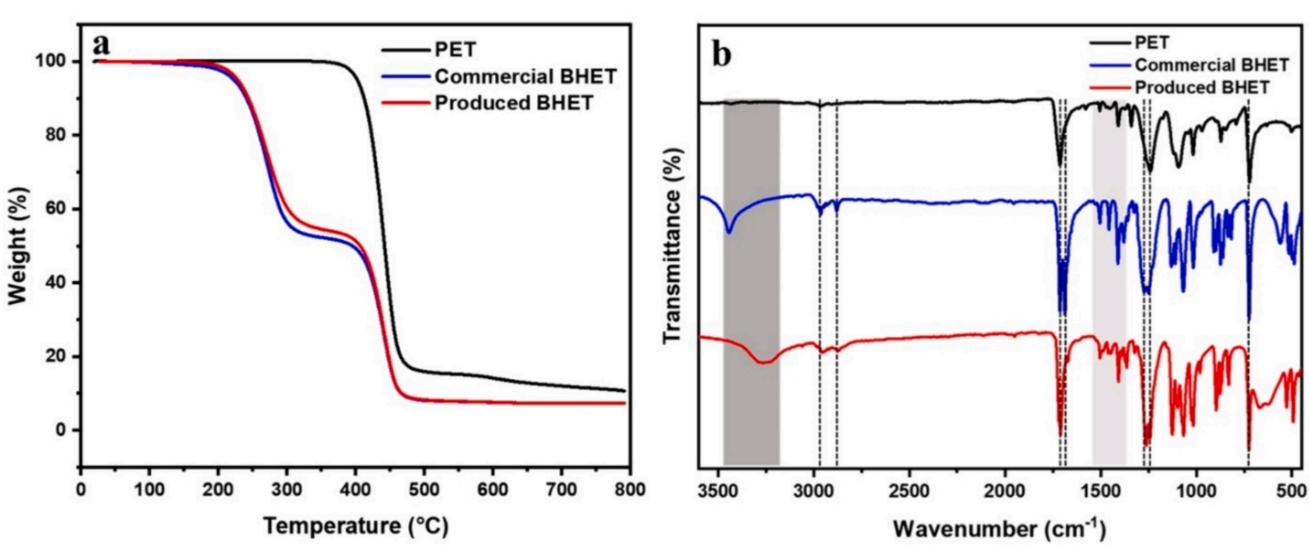


Fig. 3. Characterizing BHET: a) TGA profiles and b) FT-IR spectra.

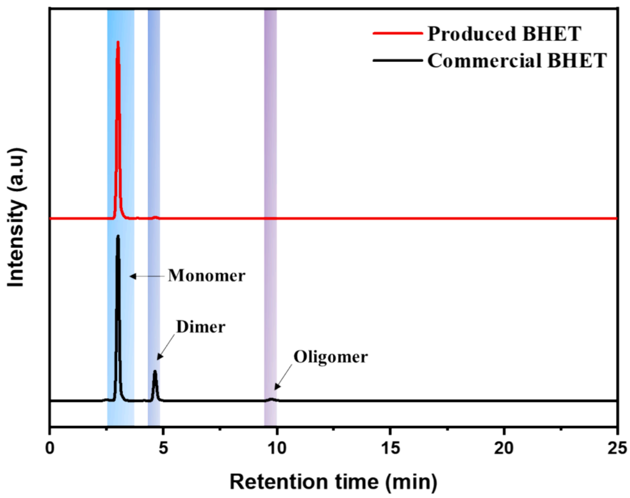


Fig. 4. HPLC traces of produced (red) and commercial (black) BHET.

adjacent to the hydroxyl (–OH) and carboxylate (–COO) groups [52]. The areas highlighted in gray identify signals that correspond to the BHET dimer, with minor signals at 3.35 and 2.51 ppm identified as residual water and DMSO, respectively. Specifically, the spectrum of the produced BHET (Fig. 5**c**) does not exhibit any unexpected by-products, further validating the effectiveness of the glycolysis process for producing high-purity BHET. This observation is consistent with the results obtained by HPLC, reinforcing the conclusion that the synthesized BHET is of comparable purity to commercial standards with no significant oligomers or other impurities present. Thus, ^{1}H NMR spectroscopy not only verified the structural composition of the produced BHET but also confirmed its high purity, thereby providing substantial evidence for the efficacy of the glycolysis method used in this study for PET recycling.

The purity of BHET was further verified using ICP-MS, the results of which are summarized in Table 4. Remarkably, ICP-MS did not detect any metal in the produced BHET. Typically, metal impurities are present at levels of approximately 5–6 ppm in BHET produced with catalysts such as Zn or Ca [53]. The absence of such impurities in our product suggests that we produced exceptionally pure BHET, which is likely due to the use of large catalyst particles that are easily separated by filtration. This finding is consistent with other analytical data, confirming that ultrapure BHET was produced by PET glycolysis in this study. Successfully removing the catalyst through simple filtration meets the research objective of using large catalyst particles to achieve pure BHET via PET glycolysis.

4. Conclusion

Herein, we successfully demonstrated the efficacy of a microsized MgO/SiO_2 catalyst for the glycolytic depolymerization of post-consumer PET, which presents a significant advancement in chemical recycling. The use of a catalyst with particles that are substantially larger than those of conventional counterparts not only simplifies the separation process, thereby enhancing the purity of the resultant BHET, but also improves the overall efficiency of the recycling process, as evidenced by the notable 95.1 % yield of BHET with minimal metal contamination. The purity of BHET exceeded 99.85 % according to the HPLC and NMR results, and no metal contaminants were detected by ICP-MS. Despite the initial assumption that larger particles may hinder catalytic activity, the catalyst developed in this study showed comparable efficiency to that of its nanosized counterparts. These findings highlight the potential of microsized catalysts for enhancing the sustainability of PET recycling. This breakthrough suggests that the overall recycling process can be optimized without compromising catalytic performance. These results emphasize the capabilities of micro-sized catalysts for addressing challenges associated with recycling PET, especially in terms of catalyst retrieval and end-product purity. Therefore, this study contributes to the broader goal of sustainable PET management by offering a viable pathway for the closed-loop recycling of this widely used plastic, and is aligned with global sustainability objectives.

CRediT authorship contribution statement

Eun Hyup Kim: Writing – original draft, Investigation. **Inseo Park:** Methodology, Formal analysis, Data curation. **SeungHwan Kim:** Formal analysis. **Jeong F. Kim:** Methodology, Investigation, Conceptualization. **Yo Han Choi:** Resources, Methodology, Conceptualization. **Hoik Lee:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

Table 4
ICP-MS data for BHET produced by MgO/SiO_2-catalyzed PET glycolysis.

Glycolysis catalyst	Metal content[a]	Reference
10 wt% MgO/SiO_2	n.d	This work
30 wt% MgO/SiO_2	n.d	This work
$ZnAc_2$	6 ppm	[53]
$CaAc_2$	5 ppm	

[a] Measured by ICP-MS; n.d = not detected.

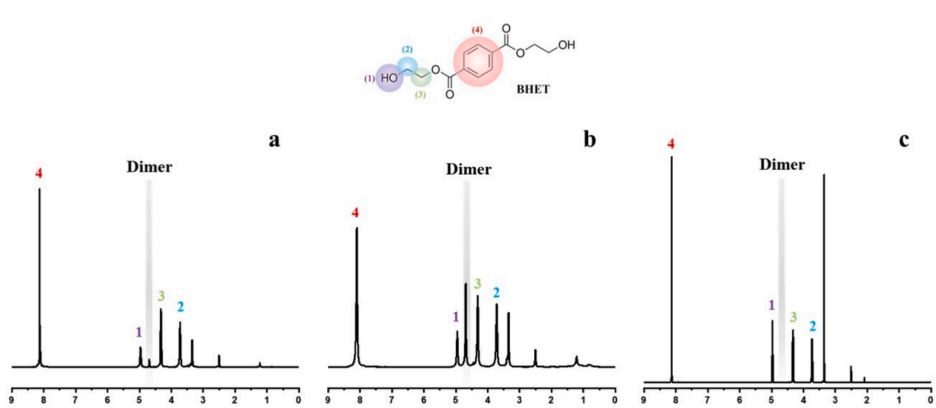


Fig. 5. ^{1}H NMR spectra of BHET: a) commercial BHET, b) BHET dimer, and c) BHET produced using the 10 wt% MgO/SiO_2 catalyst.

the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

This work was supported by the National Research Foundation of Korea (NRF) (Grant No. 2022R1F1A1074129), the Korea Institute of Industrial Technology (KITECH), and the Nano Material Technology Development Program (2021M3H4A1A04092885) through the National Research Foundation of Korea, which is funded by the Ministry of Science and ICT.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2024.155865.

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